

RANTOUL, IL, USA

WMO: 722194

Lat: 40.293N

Lon: 88.142W

Elev: 225

StdP: 98.66

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04896

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.8	-15.1	-22.2	0.5	-16.2	-19.6	0.7	-14.2	12.8	-0.2	11.8	-0.8	3.3	310	0.509

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	33.6	24.3	32.5	23.9	31.2	23.2	27.2	30.8	25.9	30.0	24.9	29.1	4.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.2	22.2	29.8	24.9	20.5	28.7	23.7	19.0	27.2	87.3	30.8	81.7	29.8	76.8	29.2	33.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	10.2	8.8	DB	-21.8	35.3	3.6	1.6	-24.4	36.5	-26.5	37.4	-28.5	38.3	-31.1	39.5	(4)
(5)			WB	-21.8	28.7	3.7	1.7	-24.4	30.0	-26.6	31.0	-28.7	32.0	-31.4	33.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	-3.1	-1.7	5.3	11.9	18.3	23.0	24.0	23.2	20.3	13.2	5.9	0.0
	DBStd	11.01	6.71	6.65	6.30	5.19	5.09	3.24	3.00	2.94	3.92	5.25	5.30	5.93
	HDD10.0	1435	409	330	174	38	3	0	0	0	0	25	142	313
	HDD18.3	3050	666	561	406	201	67	3	1	3	24	178	372	569
	CDD10.0	2078	1	3	29	94	259	391	434	411	309	123	20	3
	CDD18.3	650	0	0	2	7	65	145	177	154	82	17	1	0
	CDH23.3	6128	0	0	18	105	659	1354	1675	1366	779	171	2	0
	CDH26.7	2176	0	0	2	16	233	488	635	471	286	46	0	0
Wind	WSAvg	3.9	4.8	4.6	4.6	4.8	4.0	3.3	2.5	2.4	2.8	3.8	4.3	4.4
Precipitation	PrecAvg	1001	61	50	69	95	111	108	110	104	66	86	84	60
	PrecMax	1411	168	127	157	226	214	232	245	181	193	177	192	160
	PrecMin	799	14	8	17	39	32	26	26	55	16	34	14	28
	PrecStd	152	35	36	38	44	49	49	57	42	42	44	40	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.4	18.5	26.0	28.4	33.2	34.7	36.5	35.0	34.8	31.1	22.3	17.2
		MCWB	13.9	14.6	17.6	16.5	21.1	23.9	24.7	24.8	21.9	20.8	16.0	14.9
	2%	DB	12.1	14.7	21.3	26.4	31.4	32.7	33.1	32.7	32.4	27.6	18.8	13.6
		MCWB	10.6	12.4	15.4	15.8	20.3	24.0	25.6	24.7	22.2	18.1	14.8	12.0
	5%	DB	8.8	10.3	17.8	24.0	29.3	31.4	32.1	31.2	30.3	25.0	16.3	10.5
		MCWB	7.6	8.2	13.6	15.7	19.4	23.0	25.7	23.9	22.2	17.2	12.9	9.1
	10%	DB	5.6	7.3	15.2	21.2	27.3	29.8	30.2	29.3	27.7	22.1	14.0	7.9
		MCWB	4.1	5.4	11.7	14.7	18.9	22.0	24.0	22.8	19.5	16.5	10.6	6.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.2	15.8	18.1	20.2	24.2	27.8	29.6	28.6	25.7	23.0	18.0	15.6
		MCDB	15.5	17.7	23.5	23.9	29.7	32.1	32.1	31.9	30.9	28.1	20.9	16.6
	2%	WB	10.8	12.1	16.3	18.5	22.8	26.3	27.6	26.5	24.5	20.9	15.7	12.2
		MCDB	11.7	14.3	21.0	23.3	27.9	30.6	31.1	30.2	29.4	25.0	17.8	13.4
	5%	WB	7.5	8.7	14.6	17.0	21.7	25.0	26.2	25.3	23.2	19.4	13.5	9.2
		MCDB	8.6	10.2	17.6	21.4	26.6	29.2	30.0	29.3	27.3	22.9	16.1	10.2
	10%	WB	4.2	5.2	12.2	15.5	20.5	23.8	25.1	24.1	22.0	17.5	11.2	6.7
		MCDB	5.5	7.4	14.5	20.1	25.1	27.6	29.2	27.9	25.8	21.6	13.6	7.8
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.7	9.9	11.7	11.7	10.8	10.4	10.8	12.1	11.4	9.5	7.3
		MCDBR	11.1	12.3	13.9	15.4	14.2	12.6	11.6	12.3	14.5	14.9	13.1	10.3
		MCWBR	10.0	10.1	9.1	7.7	6.0	5.9	5.9	6.2	6.0	7.3	9.1	9.1
	5% WB	MCDBR	10.7	12.0	12.5	12.6	12.1	10.6	10.4	10.7	11.8	11.7	11.9	9.8
		MCWBR	10.0	10.5	8.7	8.3	6.6	6.2	6.6	6.5	6.0	7.2	9.4	9.0
Clear-Sky Solar Irradiance	taub	0.300	0.315	0.352	0.394	0.435	0.441	0.458	0.455	0.415	0.362	0.315	0.297	
	taud	2.432	2.383	2.365	2.252	2.194	2.221	2.155	2.152	2.243	2.388	2.476	2.453	
	Ebn at Noon	865	900	900	879	847	840	821	812	821	834	839	838	
	Edh at Noon	80	97	111	133	144	140	149	145	124	96	76	73	
All-Sky Solar Radiation	RadAvg	1.78	2.67	3.65	4.74	5.42	6.21	6.15	5.46	4.60	3.14	2.09	1.49	
	RadStd	0.20	0.26	0.28	0.36	0.46	0.39	0.32	0.33	0.34	0.33	0.20	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page